Vehicle Gateway Control System: High-Level Design (HLD)

Document ID: VH-GW-HLD-002
Version: 2.0
Document Type: Synthetic High-Level Design Knowledge-Base Document

1. Document Information and Ground-Truth Statement

1.1 Document Information

Field	Value
Document ID	VH-GW-HLD-002
Version	2.0
System Name	Vehicle Gateway Control System
Document Purpose	Define the high-level software architecture of the Vehicle Gateway Control System
Primary Communication Protocol	Controller Area Network (CAN)
Secondary Communication Technology	Automotive Ethernet

1.2 Ground-Truth Statement

This document is completely synthetic and self-contained. It is the only source of truth for the Vehicle Gateway Control System. Any fact that is not stated in this document is not specified in this HLD.

The Vehicle Gateway Control System contains exactly five primary Software Components, exactly three primary Interfaces, exactly five explicit Signals, and exactly six Port mappings.

1.3 Terminology

Term	Meaning in this HLD
Software Component	An architectural unit of the system that owns ports and responsibilities
Interface	A named contract that groups one or more signals and defines a communication pattern
Port	A connection point of a Software Component. A Provided Port (PPort) provides an interface. A Required Port (RPort) requires an interface
Signal	A named data element contained in an interface
Provider	The Software Component that provides a signal
Consumer	The Software Component that consumes a signal
Communication Protocol	The transport mechanism that carries signals between Software Components

1.4 Authoritative Sections

- Section 4 is the authoritative source for signal ownership (provider and consumer).
- Section 7 is the authoritative source for diagnostic scope boundaries.

2. System Architecture

2.1 System Overview

The Vehicle Gateway Control System is composed of exactly five primary Software Components:

1. `SensorInterface`
2. `VehicleControlComponent`
3. `BrakeControlComponent`
4. `DiagnosticManager`
5. `CommunicationManager`

No other Software Components are defined in this HLD.

The system has two functional flows:

- The primary control flow carries VehicleSpeed, EngineSpeed, and BrakeStatus from SensorInterface through VehicleControlComponent to BrakeControlComponent.
- The diagnostic flow carries DiagnosticRequest and DiagnosticResponse between CommunicationManager and DiagnosticManager.

2.2 Software Component Responsibilities

2.2.1 SensorInterface

SensorInterface is responsible for acquiring and normalizing physical vehicle sensor information. SensorInterface provides VehicleSpeed and EngineSpeed.

2.2.2 VehicleControlComponent

VehicleControlComponent processes vehicle-speed and engine-speed information and generates brake control status. VehicleControlComponent consumes VehicleSpeed and EngineSpeed. VehicleControlComponent provides BrakeStatus.

2.2.3 BrakeControlComponent

BrakeControlComponent consumes brake-related control and status information for vehicle deceleration actuation. BrakeControlComponent consumes BrakeStatus.

2.2.4 DiagnosticManager

DiagnosticManager processes diagnostic requests and generates diagnostic responses. DiagnosticManager consumes DiagnosticRequest. DiagnosticManager provides DiagnosticResponse.

2.2.5 CommunicationManager

CommunicationManager provides communication transport services for CAN and Automotive Ethernet. CommunicationManager provides DiagnosticRequest to DiagnosticManager. CommunicationManager consumes DiagnosticResponse.

3. Software Components and Ports

3.1 Software Component List

No.	Software Component	Role Summary
1	SensorInterface	Acquires and normalizes physical vehicle sensor information

No.	Software Component	Role Summary
2	VehicleControlComponent	Processes vehicle speed and engine speed and generates brake control status
3	BrakeControlComponent	Consumes brake status for vehicle deceleration actuation
4	DiagnosticManager	Processes diagnostic requests and generates diagnostic responses
5	CommunicationManager	Provides transport services for CAN and Automotive Ethernet

SensorInterface is a Software Component. It is not an Interface. The Software Component name SensorInterface must not be confused with the three Interfaces defined in Section 4.

3.2 Port Mapping

The Vehicle Gateway Control System defines exactly six ports. The mapping below is complete.

Software Component	Port	Direction	Interface
SensorInterface	VehicleDataOut	Provided (PPort)	VehicleDataInterface
VehicleControlComponent	VehicleDataIn	Required (RPort)	VehicleDataInterface
VehicleControlComponent	BrakeStatusOut	Provided (PPort)	BrakeStatusInterface
BrakeControlComponent	BrakeStatusIn	Required (RPort)	BrakeStatusInterface
DiagnosticManager	DiagnosticOut	Provided (PPort)	DiagnosticInterface
CommunicationManager	DiagnosticIn	Required (RPort)	DiagnosticInterface

3.3 Port Statements per Software Component

- SensorInterface owns one port: VehicleDataOut, a Provided Port (PPort) of VehicleDataInterface.
- VehicleControlComponent owns two ports: VehicleDataIn, a Required Port (RPort) of VehicleDataInterface, and BrakeStatusOut, a Provided Port (PPort) of BrakeStatusInterface.
- BrakeControlComponent owns one port: BrakeStatusIn, a Required Port (RPort) of BrakeStatusInterface.

- DiagnosticManager owns one port: DiagnosticOut, a Provided Port (PPort) of DiagnosticInterface.
- CommunicationManager owns one port: DiagnosticIn, a Required Port (RPort) of DiagnosticInterface.

3.4 Port Connections

- VehicleDataOut of SensorInterface is connected to VehicleDataIn of VehicleControlComponent through VehicleDataInterface.
- BrakeStatusOut of VehicleControlComponent is connected to BrakeStatusIn of BrakeControlComponent through BrakeStatusInterface.
- DiagnosticOut of DiagnosticManager is connected to DiagnosticIn of CommunicationManager through DiagnosticInterface.

The HLD defines no additional port for DiagnosticRequest. The six ports in Section 3.2 are the complete port set.

4. Interfaces and Signals

4.1 Interface Definitions

The Vehicle Gateway Control System defines exactly three primary Interfaces.

Interface	Communication Pattern	Contained Signals
VehicleDataInterface	Sender-Receiver	VehicleSpeed, EngineSpeed
BrakeStatusInterface	Sender-Receiver	BrakeStatus
DiagnosticInterface	Client-Server / Sender-Receiver hybrid	DiagnosticRequest, DiagnosticResponse

4.1.1 VehicleDataInterface

VehicleDataInterface uses the Sender-Receiver communication pattern. VehicleDataInterface contains the signals VehicleSpeed and EngineSpeed. VehicleSpeed is contained in VehicleDataInterface. EngineSpeed is contained in VehicleDataInterface.

4.1.2 BrakeStatusInterface

BrakeStatusInterface uses the Sender-Receiver communication pattern. BrakeStatusInterface contains the signal BrakeStatus.

4.1.3 DiagnosticInterface

DiagnosticInterface uses a Client-Server / Sender-Receiver hybrid communication pattern. DiagnosticInterface contains the signals DiagnosticRequest and DiagnosticResponse.

4.2 Explicit Signals

The Vehicle Gateway Control System defines exactly five explicit Signals.

Signal	Contained in Interface
VehicleSpeed	VehicleDataInterface
EngineSpeed	VehicleDataInterface
BrakeStatus	BrakeStatusInterface
DiagnosticRequest	DiagnosticInterface
DiagnosticResponse	DiagnosticInterface

4.3 Normative Signal Provider / Consumer Relationships

This section is the authoritative source for signal ownership in VH-GW-HLD-002. If any other section appears to differ, this section takes precedence.

Signal	Provider	Consumer
VehicleSpeed	SensorInterface	VehicleControlComponent
EngineSpeed	SensorInterface	VehicleControlComponent
BrakeStatus	VehicleControlComponent	BrakeControlComponent
DiagnosticRequest	CommunicationManager	DiagnosticManager
DiagnosticResponse	DiagnosticManager	CommunicationManager

The relationships are restated as direct sentences:

- SensorInterface provides VehicleSpeed to VehicleControlComponent.
- VehicleControlComponent consumes VehicleSpeed.
- SensorInterface provides EngineSpeed to VehicleControlComponent.
- VehicleControlComponent consumes EngineSpeed.
- VehicleControlComponent provides BrakeStatus to BrakeControlComponent.

- BrakeControlComponent consumes BrakeStatus.
- CommunicationManager provides DiagnosticRequest to DiagnosticManager.
- DiagnosticManager consumes DiagnosticRequest.
- DiagnosticManager provides DiagnosticResponse to CommunicationManager.
- CommunicationManager consumes DiagnosticResponse.

4.4 Signal Ownership Summary by Software Component

Software Component	Signals Provided	Signals Consumed
SensorInterface	VehicleSpeed, EngineSpeed	None
VehicleControlComponent	BrakeStatus	VehicleSpeed, EngineSpeed
BrakeControlComponent	None	BrakeStatus
DiagnosticManager	DiagnosticResponse	DiagnosticRequest
CommunicationManager	DiagnosticRequest	DiagnosticResponse

5. Communication Architecture

5.1 Primary Protocol: Controller Area Network (CAN)

Controller Area Network (CAN) is the primary communication protocol of the Vehicle Gateway Control System. CAN is the primary communication protocol for all five explicit HLD signals:

- VehicleSpeed
- EngineSpeed
- BrakeStatus
- DiagnosticRequest
- DiagnosticResponse

CAN is the primary transport mechanism for VehicleSpeed. CAN is the primary transport mechanism for EngineSpeed. CAN is the primary transport mechanism for BrakeStatus. CAN is the primary transport mechanism for DiagnosticRequest. CAN is the primary transport mechanism for DiagnosticResponse.

5.2 Secondary Technology: Automotive Ethernet

Automotive Ethernet is the secondary communication technology of the Vehicle Gateway Control System. Automotive Ethernet is used for:

- high-bandwidth service-oriented communication
- calibration
- engineering flashing

Automotive Ethernet is not the primary transport mechanism for any of the five explicit HLD signals.

5.3 Role of CommunicationManager

CommunicationManager provides communication transport services for CAN and Automotive Ethernet. CommunicationManager is a Software Component. CAN and Automotive Ethernet are Communication Protocols or technologies, not Software Components.

5.4 Protocol Summary

Item	Value
Primary Protocol	CAN
Primary Protocol Scope	All five explicit HLD signals
Secondary Technology	Automotive Ethernet
Secondary Technology Uses	High-bandwidth service-oriented communication, calibration, engineering flashing
Transport Service Provider	CommunicationManager

5.5 Out-of-Scope Communication Details

The following are not specified in this HLD:

- Ethernet frame layouts
- IP addresses
- VLAN assignments
- SOME/IP service IDs
- CAN message identifiers and signal encodings

The SOME/IP service ID is not specified in this HLD.

6. End-to-End Data Flows and Dependency Summary

6.1 Primary Control Data Flow

The primary control flow is:

SensorInterface → VehicleSpeed / EngineSpeed → VehicleControlComponent → BrakeStatus → BrakeControlComponent

1. SensorInterface acquires and normalizes vehicle sensor information.
2. SensorInterface provides VehicleSpeed and EngineSpeed to VehicleControlComponent through VehicleDataInterface.
3. VehicleControlComponent consumes VehicleSpeed and EngineSpeed.
4. VehicleControlComponent calculates vehicle control state.
5. VehicleControlComponent provides BrakeStatus to BrakeControlComponent through BrakeStatusInterface.
6. BrakeControlComponent consumes BrakeStatus for vehicle deceleration actuation.

CAN is the primary communication protocol for VehicleSpeed, EngineSpeed, and BrakeStatus.

6.2 Diagnostic Data Flow

The diagnostic flow is:

CommunicationManager → DiagnosticRequest → DiagnosticManager → DiagnosticResponse → CommunicationManager

1. CommunicationManager receives an external diagnostic request.
2. CommunicationManager provides DiagnosticRequest to DiagnosticManager through DiagnosticInterface.
3. DiagnosticManager processes the request.
4. DiagnosticManager provides DiagnosticResponse to CommunicationManager through DiagnosticInterface.
5. CommunicationManager consumes DiagnosticResponse and performs transport of the response.

CAN is the primary communication protocol for DiagnosticRequest and DiagnosticResponse.

6.3 Dependency Summary

Software Component	Depends On (Consumes From)	Provides To
SensorInterface	None	VehicleControlComponent
VehicleControlComponent	SensorInterface	BrakeControlComponent
BrakeControlComponent	VehicleControlComponent	None
DiagnosticManager	CommunicationManager	CommunicationManager
CommunicationManager	DiagnosticManager	DiagnosticManager

The dependency direction in the Provides To column and the Depends On column is derived from Section 4.3. Section 4.3 remains authoritative.

7. Diagnostic Architecture and Explicit Out-of-Scope Boundary

This section is the authoritative source for diagnostic scope boundaries in VH-GW-HLD-002.

7.1 Diagnostic Architecture

The diagnostic architecture uses two Software Components:

- DiagnosticManager processes diagnostic requests and generates diagnostic responses.
- CommunicationManager receives external diagnostic requests and transports diagnostic responses.

The two Software Components communicate through DiagnosticInterface. DiagnosticInterface uses a Client-Server / Sender-Receiver hybrid communication pattern and contains DiagnosticRequest and DiagnosticResponse.

- CommunicationManager provides DiagnosticRequest to DiagnosticManager.
- DiagnosticManager provides DiagnosticResponse to CommunicationManager.

No additional diagnostic components are defined in this HLD.

7.2 Diagnostic Protocol and Transport

The HLD specifies:

- UDS over CAN
- ISO 15765-2 as the CAN transport standard

7.3 Diagnostic Scope Boundary

Specific UDS service IDs, DIDs, ECU addresses, and NRCs are not specified in this HLD.

Diagnostic Detail	Status in this HLD
UDS service IDs	Not specified in this HLD
Data Identifiers (DIDs)	Not specified in this HLD
ECU physical addresses	Not specified in this HLD
ECU functional addresses	Not specified in this HLD
Negative Response Codes (NRCs)	Not specified in this HLD

7.4 Scope Boundary Statements

- The UDS service ID for reading a DTC is not specified in this HLD.
- The ECU address used for DiagnosticManager is not specified in this HLD.
- The NRC returned for an unsupported diagnostic request is not specified in this HLD.
- The SOME/IP service ID is not specified in this HLD.

A question about any detail listed in this section must be answered with: not specified in the HLD.

8. Architecture Constraints and RAG Traceability

8.1 Architecture Constraints

ID	Constraint
C-01	The system contains exactly five primary Software Components
C-02	The system contains exactly three primary Interfaces
C-03	The system contains exactly five explicit Signals
C-04	The system contains exactly six Port mappings
C-05	CAN is the primary communication protocol for all five explicit HLD signals

ID	Constraint
C-06	Automotive Ethernet is the secondary communication technology
C-07	Signal provider and consumer relationships are defined only in Section 4.3
C-08	Diagnostic scope boundaries are defined only in Section 7
C-09	No additional Software Components, Interfaces, or Signals are defined

8.2 RAG Traceability: Authoritative Sections

Topic	Authoritative Section
Document identity and terminology	Section 1
Software Component responsibilities	Section 2.2
Port mapping	Section 3.2
Interface definitions	Section 4.1
Signal provider and consumer relationships	Section 4.3
Primary and secondary communication technologies	Section 5
Control data flow	Section 6.1
Diagnostic data flow	Section 6.2
Diagnostic scope boundaries	Section 7

8.3 Question Traceability

Question	Answer	Evidence Section
Which component provides VehicleSpeed?	SensorInterface	4.3
Which component consumes VehicleSpeed?	VehicleControlComponent	4.3

Question	Answer	Evidence Section
What interface contains VehicleSpeed?	VehicleDataInterface	4.1.1
Which component provides BrakeStatus?	VehicleControlComponent	4.3
Which component consumes BrakeStatus?	BrakeControlComponent	4.3
Which communication protocol is primary for the explicit HLD signals?	CAN	5.1
What is the relationship between DiagnosticManager and CommunicationManager?	They communicate through DiagnosticInterface. CommunicationManager provides DiagnosticRequest to DiagnosticManager, and DiagnosticManager provides DiagnosticResponse to CommunicationManager	4.3, 7.1
Which component provides DiagnosticResponse?	DiagnosticManager	4.3
What is the UDS service ID for reading a DTC?	Not specified in the HLD	7.3, 7.4
What is the ECU address used for DiagnosticManager?	Not specified in the HLD	7.3, 7.4
What NRC is returned for an unsupported diagnostic request?	Not specified in the HLD	7.3, 7.4
What is the exact SOME/IP service ID?	Not specified in the HLD	5.5, 7.4

8.4 Document End

End of VH-GW-HLD-002, Version 2.0.